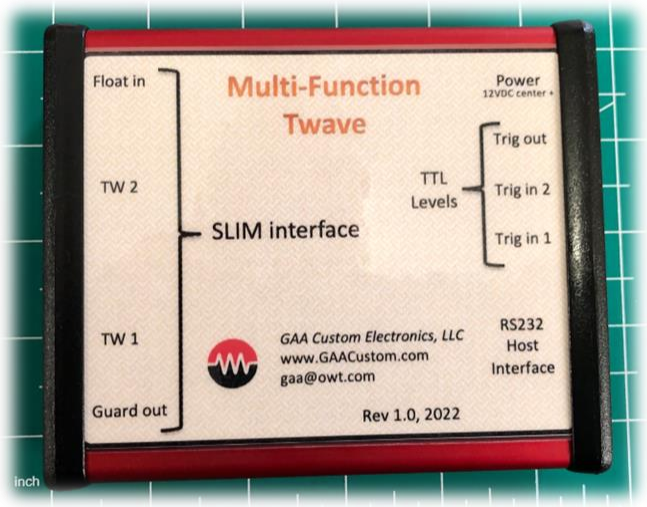
MFT

Multi-Function Twave

Operations Manual
Rev 4.0, July 9, 2024
Require firmware 1.7 or later



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WARNING

TO REDUCE THE RISK OF FIRE OR ELECTRICAL SHOCK DO NOT EXPOSE THIS EQUIPMENT TO RAIN OR MOSITURE.

WARNING

Only qualified electrical workers should operate and install this system.
Avoid spilling liquids onto/into the unit.
Do not expose to excessive heat or moisture.
Do not open – there are no user serviceable parts inside.
Do not operate in an explosive environment.

The Multi-Function Twave (MFT) is a standalone Twave driver designed to generate the traveling waves signal needed to support SLIM operations and general IMS traveling wave systems. This system generates two independent sets of Twave signals that will support advanced functions such as on-board accumulation and CRIMP. The MFT has a RS232 interface that allows the user to monitor and control the system. The features of this MFT are listed below:

- Powered with 12VDC wall supply (18 watt)
- RS232 interface with ASCII control commands, USB to RS232 cable included
- Compatible with MIPS host application
- Two separate Twave output signal sets
- User definable Twave pattern
- User adjustable voltage range
- User programable guard output
- Output voltage range of 3 to 50 volts
- Trigger output, TTL
- Two trigger inputs, TTL
- User adjustable frequency up to 25KHz
- D9 Twave connectors
- SMB digital IO, guard, and float connectors
- No power switch, the system powers up when power is applied

The MFT Traveling Wave generator enables the user to define any 8-bit binary pattern that will be used to generate 8 output waveforms for each output module. An internal adjustable clock is used to rotate this bit pattern to the left (left for forward and right for reverse directions) on each clock cycle as illustrated in the example sequence below:

00000011
00000110
00001100
00011000
00110000
01100000
11000000
10000001

At each clock pulse the pattern rotates one bit position to the left with the most significant bit rotating back to the least significant bit. Each bit position defines a waveform for a total of 8 waveforms. Each waveform is identical but shifted one clock cycle from its neighbor. These 8 waveforms appear on the 9 pin D connector.

The user can reverse the direction of the traveling wave using the provided user interface command. The user can control the pulse peak voltage (high state) from 3 to 50 volts. The output pattern can also be user defined.

Output stepping

The MFT supports output clock stepping. In this mode the output waveforms will advance by the number of clock cycles you define. Commands allow you to start and stop the clock and also define a number of clock cycles (STEPS) you wish to perform before automatically stopping the cock. The commands are listed in the host commands section of this document.

Output direction reversal

The twave signals can be set to forward direction or reverse direction. The two modules have independent control of their direction. This direction can also be set using the trigger input signals and controlled with the command string.

Phase shifting

The twave waveforms can have phase shifts applied to both forward direction and reverse direction. There are independent settings for both modules and. When the phase shifts are set to zero the two modules twave signals are in phase and the reversal will change the direction in phase with the forward direction waveform. These phase shift commands allow defining any relationship needed for your hardware design.

The system phase resolution is 45 degrees, any phase shift value in degrees can be entered and the system adjust to closest achievable value.

Output open state

The twave output channels can be set to high impedance mode using the open mask and open mode command. Separate parameters are available for module 1 and 2. The open mask is an 8-bit binary value that defines the channels that will be set to high impedance when the open mode is set. Examples:

SOMSK ,1,1111111	This will set all twave channels to high impedance for module 1
SOMSK ,1, 1	This will set twave channel 1 to high impedance for module 1

After the open mask is defined the SOPEN command can be used to apply the mask. The trigger input can also be used to apply the mask from an external event as well as the command string capability.

The open capability allows connecting two different sets of twave outputs to the same set of electrodes and using the SOPEN capability to select the active source. Additionally, one electrode could be set to high impedance and an external voltage applied for blocking or steering ions.

Command string

The MFT supports two command strings that allows you to define commands that can then be executed by an external trigger signal or by actions from the counter function described later in this document. As an example, a command string could be set to:

STWV,1,12;STWV,2,25

In this case two separate MFT command are combined, separated with a semi colon, these commands will set the TW voltage of module 1 to 12 volts and module 2 to 25 volts. The command string can be any combination of MFT commands.

The voltage and frequency setting commands allow you to define a change, let's look at another command string example:

STWV,1,+=1;SFREQ,-=1000

This command string will increase the voltage on TW module 1 by 1 volt and decrease the frequency by 1000 Hz.

The MFT commands to select a command string, set the command string, read its value and execute the command string can be found in the host commands section of this document.

Trigger inputs

The MFT has two trigger input signals. These triggers can be configured to do several different functions, additionally the trigger active edge or level can be defined. Trigger actions:

- REV1
  - Reverse the TW direction for module 1
- REV2
  - Reverse the TW direction for module 2
- OPEN1
  - Apply the open mask to module 1 outputs
- OPEN2
  - Apply the open mask to module 2 outputs
- CNT
  - Advances the internal counter
- CMD
  - Causes the execution of the command string

Counter

The MFT contains an internal event counter and that be advanced using one or more of the trigger input signals. The counters value can be read and reset by MFT commands. Additionally, the counter has a definable threshold value. When the counter's value reaches the threshold, several optional actions can occur:

- Generate a trigger output pulse
- Reset the counter value to zero
- Execute the command string

Host commands allow you to configure the counter for your application.

Clock

The MFT can generator a clock in the frequency range of 1 to 10000 Hz. This clock signal can be sent out the trigger output connector or you can use this clock to advance the counter. You will find the Clock commands later in this document.

MFT Kit Contents:

The MFT kit includes the MFT system as well as a USB to D9 communications cable, a 12 volt 18 watt wall supply, two SMB to BNC cables, and a terminator for the float input.

MFT connections:

Front panel connections

Power input

A 12VDC 18 watt wall supply has been provided with the system. Never use a supply over 18 watts. Note, there is no power switch and the system will start as soon as its powered up.

Trig out

This is a TTL level output on a SMB connector that can be used to trigger external devices. This output can have a number of functions depending the firmware version you are using in the system.

Trig in 1

Trig in 2

These are TTL level inputs on SMB connectors that allow external devices to signal the MFT to perform specific functions. These inputs can have a number of functions depending the firmware version you are using in the system.

RS232

The RS232 interface is designed to communicate with the MIPS host application. This interface has a default baud rate of 115,200. The commands are documented later in this document. Note, a D9 gender changer is attached to the MFT to allow connecting the communications cable.

Rear panel connections

Float in

The Twave and Guard output voltages are isolated and can be floated. This SMB connector allows you to apply a +/- 250V float voltage to the system. Note; this input must be grounded if you do not wish to apply a float voltage.

Guard out

This is a DC programmable voltage with a range of 3 to 50 volts. This can be used to bias the SLIM guard electrodes. This output cab be sent below 3 volts. The output is nonlinear below 3 volts and the readback value will indicate the actual output voltage.

TW 1

This is a set of 8 traveling wave output signals for TW module 1. These 8 signals produce a traveling wave rectangular shaped waveform. The output bit pattern, voltage, and

frequency can be controlled by the user. These signals are available on a 9pin D female connector with the following pin assignment:

D connector pin number	Function
1	TW phase 1
2	TW phase 2
3	TW phase 3
4	TW phase 4
5	TW phase 5
6	TW phase 6
7	TW phase 7
8	TW phase 8
9	Chassis ground

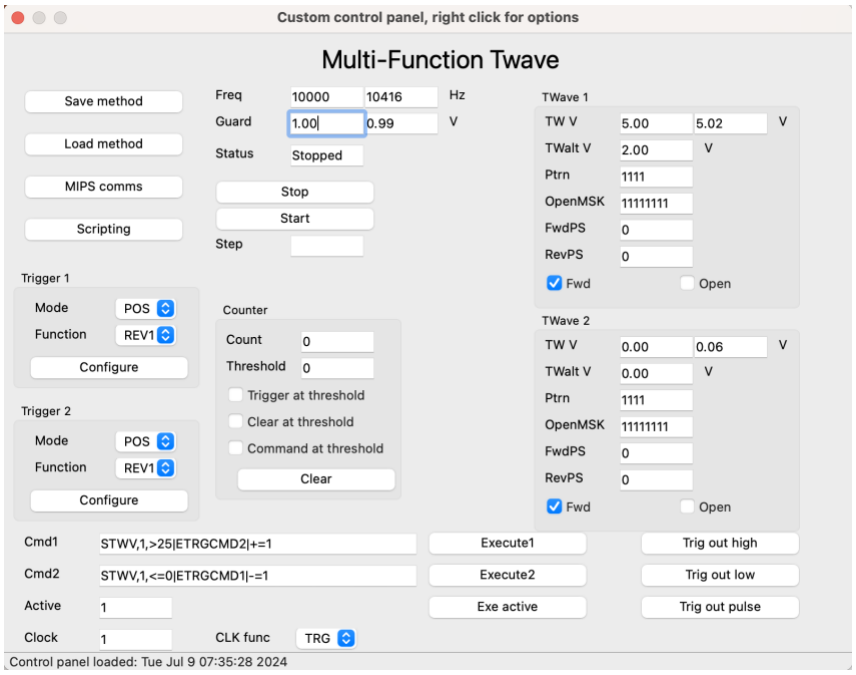
TW 2

This is a set of 8 traveling wave output signals for TW module 2. These 8 signals produce a traveling wave rectangular shaped waveform. The output bit pattern, voltage, and frequency can be controlled by the user. These signals are available on a 9pin D female connector with the following pin assignment:

D connector pin number	Function
1	TW phase 1
2	TW phase 2
3	TW phase 3
4	TW phase 4
5	TW phase 5
6	TW phase 6
7	TW phase 7
8	TW phase 8
9	Chassis ground

MIPS App Interface

Interfacing with the MIPS host application and using the control panel feature is the best way to use this system and allows monitoring and control of all the critical parameters. Below is a screen capture of the MFT's parameters displayed on a control panel:



The control panel configuration file with the imbedded scripts needed to support this interface are included with your MFT along with the files needed to install the MIPS application on MAC and PC computers. Please refer to the thumb drive that came with your MFT system.

RS232 Host Computer Interface:

All commands are ASCII text.
All commands sent to the MFT are terminated with a carriage return (CR) or line feed (LF) character.
All messages sent from the MFT are terminated with a CR and LF.
After a command is received (after the CR or LF is received) MFT will respond with an ASCII ACK (0x06) then a CR or an ASCII NAK (0x15) then a “?” character and a finally a CR. A NAK indicates the command or its arguments was not understood.
When a NAK is sent the “?” character is sent to inform users communicating with a terminal emulator that the command was not understood.

Several of the commands that allow setting system parameters also support a syntax that allows incrementing and decrementing the current value. If you enter the parameter value desired then the system will set the value as entered, if you precede the value with += then the parameter will be incremented by the value following the +=. If you precede the value with -= then the parameter will be decremented. Examples:

STWV,1,25	Sets module 1 twave voltage to 25 volts
STWV,1,+=5	Increases module 1 twave voltage by 5 volts
STWV,1,-=3	Decreases module 1 twave voltage by 3 volts

The commands that support the change syntax are identified below as support change syntax.

Additionally, these advanced commands support an if-then-else syntax. Below is an example:

STWV,1,>10|12|SCMD_2

In this example the current value of STWV,1 is compared to 10 and if its value is greater that 10 then it will be assigned the value 12, if its less than or equal to 10 then the command SCMD,2 will be executed. Note that the comma was replaced with an underscore in the command entered, this is required to properly process the command.

The following logical operations are supported:

>
>=
<
<=
=

The value after the first | character is the true condition and the value after the second | character is the false condition. These values can be a numeric value that is assigned to the parameter, an increment or decrement command or any valid MFT commands. Note you must enter the MFT commands using the underscore character in place of the comma.

General system commands:

GVER
: Returns the current version string.

GERR
: Returns an integer error code indicating the reason for the last NAK. This value is never cleared and always indicates the communications last error.

GNAME
: Returns a string that contains the name of the MFT system.

SNAME,Name
: This command will set the MFT system name. The Name value is a user defined string. Make sure you issue a Save command to save the changes to flash memory or this name will be lost when power is cycled.

SAVE
: This command will save the MFT controller parameters to the flash memory.

RESTORE
: This command will read the current settings from the flash memory.

RESET
: This command will cause the MFT control computer to reboot.

MUTE, Value

This command will stop all messages from the MFT system to the host computer. The received commands will still be processed but no response will be issued. The value parameter turns this mode on and off where:

Value = OFF to turn the responses on and ON to silence the MFT system.

ECHO,TRUE or FALSE

Setting ECHO true will cause MFT to include the command with all responses.

GCMDS

This command will cause the MFT system to display a list of valid commands that it understands.

THREADS

The MFT system uses a threading control and each module runs on its own Thread as well as a number of threads for control functions. This command displays all the threads and few key parameters. This command is used to monitor the system and check status, mainly a diagnostic tool.

STHRDENA,Thread name,TRUE or FALSE

This command allows you to enable or disable a Thread. This is done by using the thread name and defining it as TRUE or FALSE. If set to FALSE then the thread will stop. This is development and diagnostic tool.

SBAUD,baud

This command allows you to define a new baud rate, the default is 115200. This value will take affect when the system is restarted. Make sure you set a valid value or will lose the ability to communicate!

GBAUD

This command returns the current baud rate value.

GSTATUS

The command returns the status of the MFT, this is a string with the follow results:

Running
Stopped
Stepping

SENA, TRUE or FALSE

This command will allow you enable (TRUE) or disable (FALSE) the Twave outputs.

GENA

Returns the enable status, TRUE or FALSE

SFREQ, freq

This command sets the frequency of the Twave outputs in Hz. This is the frequency of each channel of a module. Note that both Twave channel must operate at the same frequency. This command supports change syntax and if-then-else.

GFREQ

Returns the requested Twave frequency in Hz.

GAFREQ

Returns the actual Twave frequency in Hz. Internal counters are used to generate the frequency and this returned value indicate the actual frequency that is generated based on the users requested frequency.

SFWD, module, TRUE or FALSE

The traveling waveform can have a forward or revered direction. This command allows you to set the direction of each module. Valid modules number are 1 or 2. TRUE indicates forward and FALSE indicate reverse direction.

GFWD, module

Returns the select module's direction, TRUE is forward.

SPTRN, module, pattern

This command allows the user to define the output pattern that is used to generate the traveling waveform. This is a binary string that defines the output states such as 1111 indicating the 4 MSB are low and 4 LSB are high. The user can set independent patterns for the two TW modules, 1 and 2.

GPTRN, module

Return the selected module TW bit pattern.

START

Starts the TW generator clock and thus starts the waveform generation.

STOP

Stops the TW generator clock and thus stops the waveform generation.

STEP, steps

This command will start the TW generator for a user defined number of steps and then stop the clock. This allows the user to define a number of steps to move the TW.

STWV, module, voltage

This command will set the selected TW module's output peak voltage. The voltage range is 3 to 50 volts. The two modules can have independent values. This command supports change syntax and if-then-else.

GTWV, module

This command returns the selected module's requested TW voltage.

GTWVA, module

This command returns the selected module's actual measured TW voltage.

SGRD, voltage

This command allows the user to set the guard voltage, valid range is 3 to 50 volts. This command supports change syntax and if-then-else.

SGRDA, voltage

This command allows the user to set the guard voltage, valid range is 0 to 50 volts. This command supports change syntax and if-then-else. This command uses the readback value to correct for the output amplifier's nonlinearity below 3 volts. This command's response time can be several milliseconds due to the correct algorithm.

GGRD

This command returns the user requested guard voltage.

GGRDA

This command returns the actual measured guard output voltage.

SOPEN, module, TRUE or FALSE

This command will apply the open mask to the selected module, 1 or 2, if set to TRUE or disable the mask if set to FALSE. The mask is an 8-bit binary value where bits 0 thru 7 will open the TW outputs channels 1 thru 8 if the corresponding bit is set to 1. Open will set the output impedance high for the select channel.

GOPEN, module

This command will return the state of the selected module's open mask, TRUE or FALSE.

SOMSK, module, mask

This command allows the user to define the open mask for module 1 or 2. The mask value is an 8-bit binary value. If the value is set to 11111111 then when applied all 8 channels of the selected module will enter the open or high impedance state,

GOMSK, module

This command returns the binary value of the selected module open mask.

SFWDPS, module, phase

This command allows the user to define a phase shift, for the forward direction, in degrees for the selected module, 1 or 2. The system phase resolution is 45 degrees so valid values are multiples of 45 both positive or negative values are valid. The system will adjust input phase shift values to the nearest valid value. This command supports change syntax.

GFWDPS, module

This command will return the forward direction phase shift, in degrees, for the selected module.

SREVPS, module, phase

This command allows the user to define a phase shift, for the reverse direction, in degrees for the selected module, 1 or 2. The system phase resolution is 45 degrees so valid values are multiples of 45 both positive or negative values are valid. The system will adjust input phase shift values to the nearest valid value. This command supports change syntax.

GREVPS, module

This command will return the reverse direction phase shift, in degrees, for the selected module.

STRGCMD1, commands

This command allows the user to define a set of MFT commands that can be executed by external events such as a trigger input. The series of commands are delimited or separated using a semi-colon. This command saves the command in command string 1.

GTRGCMD1

This command will return the current command string 1.

STRGCMD2, commands

This command allows the user to define a set of MFT commands that can be executed by external events such as a trigger input. The series of commands are delimited or separated using a semi-colon. This command saves the command in command string 2.

GTRGCMD2

This command will return the current command string 2.

ETRGCMD

This command will execute the active command.

ETRGCMD1

This command will execute the command string 1 and set the active command string to 1.

ETRGCMD2

This command will execute the command string 2 and set the active command string to 2.

SCMD, value

This command allows you to define the active command string, value can be set to 1 or 2.

GCMD

This command returns the active command string number, 1 or 2.

MFT Trigger input and output commands:

TRIG1, mode, function

TRIG2, mode, function

These commands allow the user to define the actions that will occur when a trigger signal is received on trigger 1 or 2 inputs to the MFT. The mode value defines the edge or level of the trigger signal that will signal action, value values:

POS

Positive edge or active high level

NEG

Negative edge or active low level

CHANGE

Positive or negative edge

NA

Disables the trigger

The function parameter defines the action the trigger will perform, valid values are:

REV1

Sets module 1 direction to reverse when the trigger is active

REV2

Sets module 2 direction to reverse when the trigger is active

OPEN1

Sets module 1 open state when the trigger is active

OPEN2

Sets module 2 open state when the trigger is active

CNT

Increments the counter value on the active edge of the trigger

CMD

Executes the command string on the active edge of the trigger

GTRIGIN, trigger

This command will read the selected trigger's state. The trigger value can be 1 or 2 and the returned value is 0 or 1.

TRIGOUT, value

This command allows you to control the Trig out TTL signal. The value is a string and the valid options are listed below:

HIGH	Sets the output to logic level 1, or 5 volts
LOW	Sets the output to logic level 0, or 0 volts
PULSE	Generates a 1mS pulse on the output. If the output is set to LOW this pulse will be active high and if the output is set HIGH this will generate an active low pulse

MFT range limit commands:

SMAX,type,channel,value

This command will set the maximum value limit for the parameter defined by type. The channel parameter is optional and only entered for limit types that require a channel. The value parameter is the user programable limit.

Type	= TWV, GRD, FREQ
Channel	= 1 or 2 and only required for type TWV
Value	= Programmable limit

GMAX,type,channel

This command will return the maximum value limit for the parameter defined by type. The channel parameter is optional and only entered for limit types that require a channel.

Type	= TWV, GRD, FREQ
Channel	= 1 or 2 and only required for type TWV

SMIN,type,channel,value

This command will set the minimum value limit for the parameter defined by type. The channel parameter is optional and only entered for limit types that require a channel. The value parameter is the user programable limit.

Type = TWV, GRD, FREQ
Channel = 1 or 2 and only required for type TWV
Value = Programmable limit

GMIN,type,channel

This command will return the minimum value limit for the parameter defined by type. The channel parameter is optional and only entered for limit types that require a channel.

Type = TWV, GRD, FREQ
Channel = 1 or 2 and only required for type TWV

MFT clock commands:

SCLOCK, freq

This command sets the internal clock frequency, freq is the frequency in Hz. The valid range is 0 to 10000Hz, 0 will disable the clock.

GCLOCK

This command returns the current clock frequency in Hz.

SCLKFUN, function

This command allows you to define the internal clock function, function can be set to:

NA, clock will do nothing
TRG, clock will be output on the Trigger out connector
CNT, Clock will advance the internal counter

GCLKFUN

This command returns the internal clock function.

MFT counter commands:

GCNT

This command will return the current value of the counter. The counter is a 32 bit counter.

CLRCNT

This command will reset the counter value to 0.

SCNTTRG, count

This command sets a threshold value for the counter, the threshold valid will trigger the counter to take action when the threshold is reached.

GCNTTRG

Return the current threshold value set for the counter.

STRIGCNT, TRUE or FALSE

This command will set the trigger generation flag to TRUE or FALSE. If this flag is set to TRUE then the counter will generate an output trigger when the count value reaches the threshold.

GTRIGCNT

Returns the state, TRUE or FALSE, of the counter trigger generation flag.

STRIGRST, TRUE or FALSE

This command will set the counter reset flag to TRUE or FALSE. If this flag is set to TRUE then the counter will reset its count value to zero when the count value reaches the threshold.

GTRIGRST

Returns the state, TRUE or FALSE, of the counter reset flag.

STRIGCMD, TRUE or FALSE

This command will set the command execution flag to TRUE or FALSE. If this flag is set to TRUE then the counter will execute the command string when the count value reaches the threshold.

GTRIGCMD

Returns the state, TRUE or FALSE, of the command execution flag.

Warrantee

GAA Custom Electronics, LLC warrantees the MIPS system to be free from defects in materials and workmanship and will repair or replace the unit for a period of one year. This warrantee assumes the system is operated in compliance with the procedures and recommendation outlines in this document. GAA Custom Electronics, LLC will also provide free phone support and firmware bug fixes for up to one year. The addition of new features is not covered in this warrantee.

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